

AVIATION CUSTOMER EXPERIENCE & SERVICE ANALYTICS

1. Project Overview and Objective

This project focuses on analyzing airline customer reviews to understand customer satisfaction, service performance, recommendation patterns, and trends in airline customer experience. Using the **Skytrax Airline Review Data**, the project applies Python-based data cleaning, preprocessing, exploratory data analysis, and visualization to transform customer review data into meaningful analytical insights.

The analysis focuses on airline reviews from **2021–2025** to provide insights into recent customer experiences. The cleaned and analyzed data will subsequently be used for Power BI dashboard development to present key customer satisfaction and service performance metrics. The primary objective is to identify patterns in customer ratings, evaluate different service dimensions, compare customer experiences across airlines and traveller categories, and identify service areas that may influence overall satisfaction and recommendation behaviour.

2. Data Sources

- **Source Description and Timeline:** Airline Travel Reviews Data (Skytrax), covering airline customer reviews from **2015–2025**.
 - **Original Dataset Source:** Mendeley Data – Airline Travel Reviews Data (Skytrax)
 - **Dataset Used for Analysis:** Reviews from **2021–2025**
 - **Analysis Dataset Size:** **21,984 records × 19 features**
 - **Domain:** Aviation / Airline Customer Experience
 - **Dataset Format:** CSV
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3. Problem Statement

- To analyze airline customer reviews across different airlines and traveller categories to identify patterns in customer satisfaction and service performance.
- To evaluate customer ratings across key service dimensions such as seat comfort, cabin staff service, food and beverages, ground service, inflight entertainment, Wi-Fi/connectivity, and value for money.
- To analyze recommendation patterns and understand how overall ratings and service experiences vary across different customer segments.
- To identify trends in airline customer experience from **2021–2025** and highlight service areas that may require improvement.

- To compare customer experiences across airlines, traveller types, and seat types to generate meaningful business insights.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
review_date	Date	Date on which the customer review was posted
over_all_rating	Numeric (Integer)	Overall customer rating given to the airline on a scale of 1–10
review_header	String (Text)	Title or heading of the customer review
trip_verified	Categorical	Indicates whether the review is verified
review_text	String (Text)	Detailed textual feedback provided by the customer
aircraft	String (Text)	Aircraft information associated with the review; excluded during preprocessing due to high missing and inconsistent values
type_of_traveller	Categorical	Type of traveller, such as Business, Couple Leisure, Family Leisure, or Solo Leisure
seat_type	Categorical	Type or class of seat used by the customer
route	String (Text)	Flight route mentioned in the review
date_flown	Date / Text	Date or period when the flight was taken
seat_comfort	Numeric	Customer rating for seat comfort on a 1–5 scale
cabin_staff_service	Numeric	Customer rating for cabin staff service on a 1–5 scale
food__beverages	Numeric	Customer rating for food and beverage services on a 1–5 scale
inflight_entertainment	Numeric	Customer rating for in-flight entertainment on a 1–5 scale
ground_service	Numeric	Customer rating for ground services on a 1–5 scale
wifi__connectivity	Numeric	Customer rating for Wi-Fi and connectivity services on a 1–5 scale
value_for_money	Numeric	Customer rating for value for money on a 1–5 scale
recommended	Categorical	Indicates whether the customer recommends the airline
airline_name	Categorical	Name of the airline reviewed

5. Tools & Technologies

- **Python:** Data loading, cleaning, transformation, exploratory data analysis, and visualization.
- **Pandas:** Data manipulation, missing-value analysis, filtering, and transformation.
- **NumPy:** Numerical operations and statistical calculations.
- **Matplotlib:** Data visualization and customized charts.
- **Seaborn:** Statistical visualization and exploratory analysis.

- **Power BI:** Subsequent data modelling, DAX calculations, visualization, and interactive dashboard development.
 - **Google Collab:** Python development and notebook-based analysis.
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6. Data Pre-Processing - EDA

Tasks Performed:

- **Import Libraries** – Import Pandas, NumPy, Matplotlib, Seaborn, and Warnings.
- **View First 5 Rows** – Use `df.head()` to understand the data.
- **Check Dataset Size** – Use `df.shape` to check rows and columns.
- **Check Data Types** – Use `df.info()` to understand column types and structure.
- **Statistical Summary** – Use `df.describe()` to view numerical statistics.
- **Check Missing Values** – Use `df.isnull().sum()` to identify missing values.
- **Check Duplicate Records** – Use `df.duplicated().sum()` to identify duplicates.

Summary of Complete Data Cleaning:

- **Missing Values:** Handled missing categorical values using Unknown; retained missing rating values as NaN.
 - **Duplicates:** Checked and found **no duplicate records**.
 - **Data Types:** Converted `review_date` and `date_flown` to datetime format.
 - **Categorical Values:** Standardized inconsistent values in categorical columns.
 - **Derived Columns:** Created `review_year` and `rating_category` from `review_date` and `over_all_rating`.
 - **Unnecessary Column:** Removed `aircraft` column due to high missing values.
 - **Data Validation:** Verified rating ranges and found no invalid values.
 - **Final Dataset:** **21,984 records × 20 columns**.
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7. Python Visualizations

- Created basic visualizations using Matplotlib and Seaborn to understand customer ratings and service performance.
- Used bar charts to compare ratings across traveller types, seat types, and service areas.
- Used count plots to analyze customer recommendation patterns and categorical distributions.
- Used histograms/distribution plots to examine the distribution of overall customer ratings.
- Used line charts to analyze customer satisfaction trends from 2021–2025.

- Used correlation heatmaps to identify relationships between overall ratings and individual service factors.
 - Used these visualizations as an initial exploratory analysis before developing the interactive Power BI dashboard.
 - These Python visualizations provided an initial understanding of customer satisfaction, service performance, recommendation behaviour, and trends, which guided the selection of KPIs and visualizations for the final Power BI dashboard.
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8. Dashboard Development (Power BI)

An interactive Power BI dashboard was developed to transform the analyzed airline review data into meaningful and user-friendly visual insights.

Dashboard Features:

- Created an interactive two-page dashboard:
 - **Page 1 – Aviation Customer Experience Overview**
 - **Page 2 – Service and Customer Experience Drivers**
- Used KPI Cards, Column Charts, Bar Charts, Treemap, Line Chart, Scatter Chart, Matrix Heatmap, and Tables.
- Implemented interactive Slicers for:
 - Review Year
 - Airline
 - Seat Type
 - Traveller Type
 - Recommendation
- Added tooltips to provide additional airline-level information.
- Applied conditional formatting to the service-performance matrix to highlight high and low ratings.
- Used interactive filtering to allow users to explore customer experience across different segments.
- Designed the dashboard with a consistent aviation-themed layout for clear presentation and storytelling.

Calculated Columns & DAX Measures:

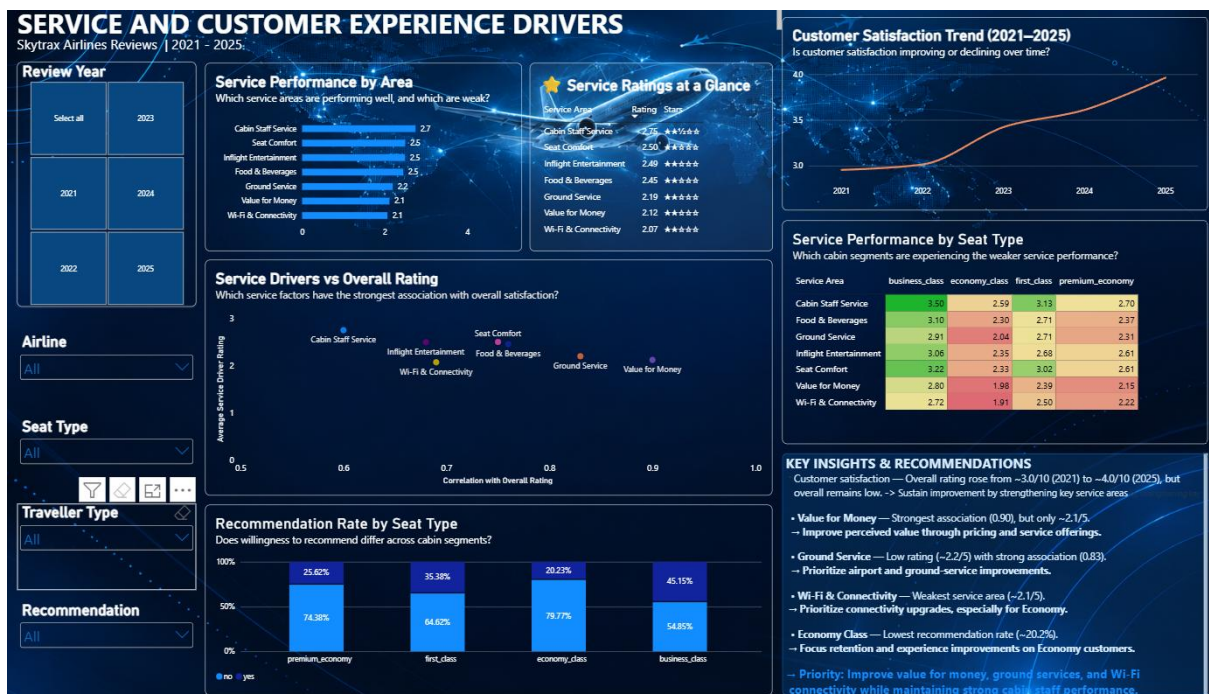
Developed DAX measures to calculate key customer experience KPIs, service performance, recommendation metrics, and airline-level insights.

List of DAX Measures Used:

Measure Name	Purpose
Total Reviews	Counts the total number of customer reviews
Average Overall Rating	Calculates the average overall customer rating
Recommended Customers	Counts customers who recommended the airline
Recommendation Rate	Calculates the percentage of customers recommending the airline
Airlines Reviewed	Counts the number of unique airlines reviewed
Average Service Rating	Calculates the average rating across all service areas
Average Service Area Rating	Calculates the average rating for each individual service area
Average Service Driver Rating	Calculates the average rating for service factors used in driver analysis
Service Rating Stars	Converts service ratings into a visual star representation

Supporting DAX Tables/Columns:

- **Service Areas** – Used to dynamically analyze individual service categories.
- **Service Drivers** – Used to compare service ratings against their correlation with overall satisfaction.
- **Seat Type Sort** – Used to maintain the desired order of cabin segments in visualizations.



9. Insight Generation and Recommendations

The analysis of Skytrax airline reviews identified important patterns in **customer satisfaction**, **service performance**, **recommendation behaviour**, and **cabin-level experience**.

Key Findings

- Overall customer satisfaction is relatively low, with an average overall rating of approximately **3.3/10**.
- Only **24.4%** of customers recommend the airline.
- Customer satisfaction shows an **improving trend from 2021 to 2025**, although the overall rating remains relatively low.

- **Value for Money** has the strongest association with overall satisfaction (**0.90**).
- **Ground Service** also shows a strong association with overall satisfaction (**0.83**).
- **Wi-Fi & Connectivity** has the lowest average service rating at approximately **2.1/5**.
- **Economy Class** generally shows weaker service performance and has the lowest recommendation rate.
- **Business Class** records stronger overall and service ratings compared with other cabin segments.
- **Cabin Staff Service** is the highest-rated service area at approximately **2.75/5**.

Analysis Insights

Descriptive

- Analyzed overall customer satisfaction and rating distribution.
- Identified customer recommendation patterns.
- Compared satisfaction across airlines, traveller types, and seat types.
- Evaluated the performance of individual service areas.
- Examined customer satisfaction trends from 2021–2025.

Diagnostic

- Investigated service factors associated with overall customer satisfaction.
- Identified **Value for Money** and **Ground Service** as strong satisfaction-associated factors.
- Compared service performance across cabin segments.
- Identified **Economy Class** as an important area requiring improvement.

Predictive

Historical satisfaction trends can help airlines monitor future customer experience patterns. However, this project focuses primarily on **descriptive and diagnostic analysis rather than predictive modelling**.

Prescriptive

- Airlines should improve **perceived value for money**, particularly for Economy customers.
- Ground-service processes should be improved to enhance customer experience.
- Airlines should prioritize improvements in **Wi-Fi and connectivity**.
- Successful service practices from higher-performing cabin segments can be considered for improving other passenger segments.
- Customer feedback should be continuously monitored to sustain the improving satisfaction trend.

Recommendations

Based on the analysis:

1. **Improve Value for Money** – Strengthen the overall value proposition, particularly for Economy passengers.
2. **Improve Ground Services** – Focus on airport support, baggage handling, and service responsiveness.
3. **Enhance Wi-Fi Connectivity** – Improve reliability and availability of onboard connectivity.
4. **Focus on Economy Experience** – Address key service gaps affecting Economy passengers.
5. **Maintain Strong Cabin Staff Performance** – Continue practices contributing to positive cabin-service experiences.

10. Conclusions

The project transformed Skytrax airline review data into meaningful customer experience insights using Python and Power BI. The analysis identified key satisfaction drivers, service gaps, and differences across passenger segments. The interactive dashboard enables users to explore these patterns and supports data-driven recommendations for improving airline customer experience.